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Long-Distance Transport of *Puccinia striiformis* f. sp. *tritici* by Upper Airflow on the Yunnan–Guizhou Plateau Disrupts the Balance of Agricultural Ecology in Central China

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Abstract

Long-distance dispersal of plant pathogens in the air can establish diseases in other areas and lead to an increased risk of large-scale epidemics. Wheat stripe rust, caused by *Puccinia striiformis* f. sp. *tritici* (*Pst*), is one of the most destructive diseases of wheat in China. Hubei is an important overwintering region for *Pst* in China, and this overwintering region is a determinant of stripe rust severity in eastern China. In 2017, stripe rust disease caused a pandemic in the Hubei region and resulted in great yield losses of wheat. To explain the disease pandemic, a total of 595 single-lesion samples of stripe rust were collected in spring, including 204 in five provinces in 2017 and 391 in four provinces in 2018, and genotyped with 13 simple sequence repeat makers. The samples were classified into 12 subpopulations based on the locations and year of collection. Genetic diversity was determined for the collection and each subpopulation. Differentiation and gene flow were determined between subpopulations. STRUCTURE analyses and discriminant analysis of principal components were conducted, and the results were used to infer the relationships between subpopulations. Our study revealed a new route of *Pst* transmission from the Yunnan–Guizhou Plateau to the Hubei region. The *Pst* inoculum of northwestern Hubei came from Gansu in the northwest, whereas the inoculum in southern Hubei came from the Yunnan–Guizhou Plateau via upper airflow. After the initial inocula infected wheat plants and multiplied in northern and southern Hubei, urediniospores produced in these regions further spread together along the middle reach of Hanshui Valley and made exchanges there. The finding of the new transmission route of *Pst* is important for improving integrated stripe rust disease management, which should have a profound impact on the balance of agricultural ecology in China.

Keywords: agricultural ecology, Hubei region, long-distance migration, *Puccinia striiformis* f. sp. *tritici*, wheat stripe rust, Yunnan–Guizhou Plateau

The sustained production increase of wheat, the third-largest staple cereal in China, has made important contributions to domestic food security (He et al. 2018). Wheat is planted in almost all provinces in China. The Huang–Huai–Hai region is the largest wheat-producing area in China, accounting for >60% of national production (Wan et al. 2007). However, it is constantly threatened by biotic and abiotic stresses every year. Stripe (yellow) rust, caused by *Puccinia striiformis* Westend. f. sp. *tritici* Erikss. (*Pst*), is an important fungal disease. With the long-distance dispersal of the *Pst* pathogens, the infection may occur thousands of kilometers away from the initial infected areas, which makes stripe rust one of the most destructive diseases of wheat in the world, especially in China (Chen et al. 2014; Zeng and Luo 2006). China has >20 million hectares of wheat affected by stripe rust annually. The most destructive epidemics occurred in 1950, 1964, 1990, 2002, and 2017, causing yield losses exceeding 6.0, 3.2, 1.8, 1.3, and 1.5 million metric tons, respectively (Huang et al. 2020). Growing disease-resistant varieties is considered the most economical, effective, and environmentally friendly measure for controlling wheat stripe rust (Chen et al. 2009). However, because of the rapid changes in the pathogen races, varieties may lose their resistance a few years after their release. The mechanisms causing pathogen variability include mutation, parasexuality, selection, and somatic recombination (Duan et al. 2010; Hovmøller et al. 2011; Mboup et al. 2009). When *Berberis* spp. were found to be alternative hosts for *Pst* (Jin et al. 2010), sexual recombination became a new focus of research on the causes of genetic variations (Tian et al. 2016; Wang et al. 2016). *Pst* is transmitted by airflow and then causes disease epidemics. The magnitude of the epidemics varies from year to year with variations in weather conditions; therefore, it is particularly important to continuously and thoroughly investigate genetic variations in the pathogen population and epidemic patterns to improve the management of stripe rust.

Based on intensive research on wheat stripe rust in China, including large-scale field surveys and fixed-point monitoring, combined with geographic, climatic, race, and molecular characterizations, China has been divided into three epidemiological regions: oversummering regions (autumn sources of *Pst* inoculum), overwintering regions (spring sources of *Pst* inoculum), and spring disease epidemic regions. The areas southeast of Gansu, southwest of Shaanxi, and northwest of Sichuan are considered the most important oversummering regions and the origin of new races of *Pst* in China (Chen et al. 2013). *Pst* can oversummer as the uredinial stage on volunteer wheat plants at high altitudes in mountainous regions and then disperse to wheat crops in low-altitude regions and produce clonal urediniospores that act as the inoculum source for other regions. Urediniospores can propagate continuously on winter wheat

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in overwintering regions, producing more urediniospores. The outbreaks of the disease occur in spring epidemic regions (including north of Henan, Anhui, Jiangsu, Shandong, Hebei, Shanxi, and north of Shaanxi), as well as the oversummering and overwintering regions, leading to large-scale epidemics that can cause serious economic losses including grain yield losses and cost of fungicide applications.

The *Pst* pathogen can oversummer at high altitudes in the Yunnan–Guizhou region (Shichen and Yun 1986). Nevertheless, the role of oversummer inoculum from the Yunnan–Guizhou Plateau was previously underestimated by most plant epidemiologists (Chen et al. 2013; Zeng and Luo 2006). As an important wheat planting region of southwestern China, the Yunnan–Guizhou Plateau is situated east of Nepal and the Himalayas. The terrain of the plateau changes from high to low in the southwest to northeast direction, and the climate in this region is cool. Wheat can grow in multiple seasons, and *Pst* can survive year-round because of the suitable environment and presence of host plants. *Berberis* spp. (*Pst* alternative hosts) are distributed in the Yunnan–Guizhou region, and *Pst*-susceptible *Berberis* species have been identified under controlled conditions (Li et al. 2021b). *Pst* sexual reproduction has been inferred to exist in the Yunnan–Guizhou region. The ancestral haplotype and a high mutation rate have been identified in the Yunnan *Pst* population. A recent study suggested that Yunnan is the probable origin center for *Pst* in China, and the initial inoculum from Gansu in the northwest region could have been transferred from Yunnan by upper airflow (Li et al. 2021a).

It is generally accepted that northwestern China is the most important oversummering area for *Pst* in China and is the major source of inoculum for wheat infection in autumn (Chen et al. 2014). Urediniospores from northwestern oversummering regions could be dispersed to the Sichuan Basin and northwestern Hubei region in autumn, propagate in this region in the winter, and act as the major spring inoculum source of stripe rust. According to a field survey in 2017, the total area affected by wheat stripe rust in China reached 5.56 million ha, of which 4.11 million ha was in the Huang–Huai–Huai region alone. Analysis of the *Pst* population structure in the Huang–Huai region in 2017 suggested that the pathogen migrated northeastward from Hubei to Shandong in eastern China (Wang et al. 2020). However, the disease incidence on autumn wheat at the seedling stage in northwestern wheat areas such as Gansu and Shaanxi was the lowest in the past 10 years. The number of diseased foci/ha was 1.5 to 9.0 (average 7.5), with an incidence of 0.01 to 1.00% (average 0.30%), and the number of diseased foci/ha at the reviving and jointing stages was 3.0 to 15.0 (average 4.5), with an incidence of 0.01 to 18% (average 3.60%) (Huang et al. 2018a, b). The oversummer inoculum source of Gansu could not cause the large-scale epidemics in China in that particular year (Huang et al. 2018b). Meanwhile, the disease was more severe in the southwest of China. The number of diseased foci/ha was 63.0 at the autumn seedling stage in Guizhou; furthermore, the average incidence was 2.20 to 4.60% at the reviving and jointing stages in southwestern wheat regions, and the field with the most serious epidemic showed an infection rate of 16.2% (Huang et al. 2018a, b). This finding led us to speculate that the inoculum sources responsible for the wheat stripe rust epidemic in Hubei in 2017 might be from the southwest instead of the northwest region. Therefore, the main objective of the present study was to explore the possibility that *Pst* in the central wheat areas originates from the Yunnan–Guizhou region.

Materials and Methods

Sample collection and DNA preparation. A total of 595 samples from five provinces (Yunnan, Guizhou, Sichuan, Gansu, and Hubei) were collected in the spring of 2017 and 2018. In the target provinces, samples were collected from more than three fields (Fig. 1; Supplementary Table S1). Leaves with a single fresh lesion of stripe rust and no symptoms of other diseases were sampled, and each leaf was enclosed in an envelope to avoid cross-contamination.

DNA was extracted according to a modified cetyltrimethylammonium bromide method (Ali et al. 2011). The procedure was as follows: Leaf tissues with single lesions were cut from collected samples, and each of them was transferred to a 2-ml centrifuge tube with 0.4 g (Φ = 1 mm) and 0.2 g (Φ = 0.1 mm) glass beads and then ground in a FastPrep 24 shaker (MP Biomedicals, Irvine, CA) for 1.5 min at a speed of 6.5 m/s. Then, 660 µl of 1.1% cetyltrimethylammonium bromide (pH = 8.0) and 10 µl of 20 mg/ml proteinase K were added, and the solution was mixed gently and incubated

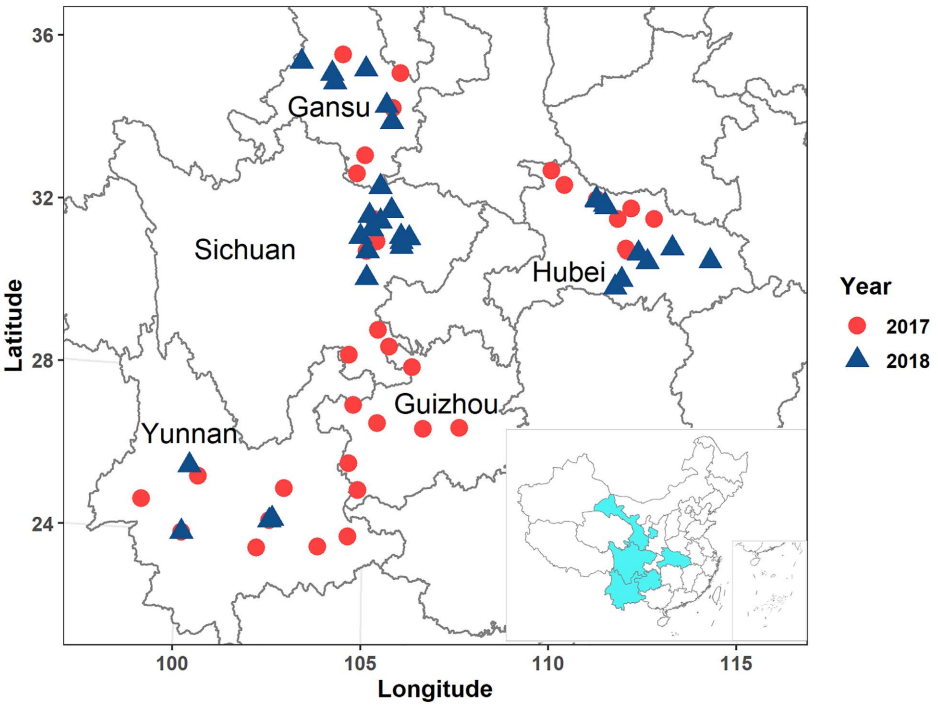

Fig. 1. Stripe rust sampling locations in Gansu, Sichuan, Yunnan, Guizhou, and Hubei provinces of China. The red circles indicate the samples collected in 2017, and the blue triangles indicate the samples collected in 2018.

in a water bath at 65°C for 1 h (the tubes were gently mixed every 15 min). Then, 660 µl of phenol/chloroform/isoamyl (25:24:1) was added, and the mixture was inverted until totally blended. The samples were centrifuged by a high-speed refrigerated centrifuge (Eppendorf, Germany) at 12,000 rpm for 10 min at 4°C. The aqueous phase was transferred to a new tube, and an equal volume of chloroform/isoamyl alcohol (24:1) was added for extraction. The supernatant was transferred to a 1.5-ml tube, and 5 µl of 100 µg/ml RNA enzyme was added. The mixture was placed in water at 37°C for 1 h and then extracted twice with the same volume of chloroform/isoamyl (24:1). A three-fifths volume of isopropanol precooled in a refrigerator was added to the tube and then kept at –20°C overnight. Subsequently, the samples were centrifuged for 10 min at 4°C and 12,000 rpm, the supernatant was discarded, and the sediment was rinsed twice with 1,000 µl of precooled 70% ethanol. The DNA was dissolved in 30 µl of double-distilled water at 4°C after drying, and the concentration of the DNA was measured by a NanoDrop spectrophotometer (ND-1000; Thermo Fisher Scientific, Waltham, MA). The DNA samples were diluted to 50 ng/µl and stored at –20°C for further use.

Simple sequence repeat genotyping. Initially, 32 primers were selected for the preliminary experiment, and 13 primers with polymorphisms were used for the subsequent experiment. PCR was carried out in a total volume of 25 µl, including 1 µl of prepared DNA, 12.5 µl of 2× Unique Taq Super Master Mix (Novogene, Beijing, China), 9.5 µl of sterile double-distilled water, and 1 µl of each forward and reverse primer. The conditions were as follows: initial denaturation at 94°C for 5 min; 35 cycles at 94°C for 30 s, 56°C for 30 s (according to the annealing temperature of the primers), and 72°C for 45 s; and a final extension at 72°C for 10 min.

The amplification products were diluted at a ratio of 1:50 after analysis in 2% agarose gel. Then, 1 µl of the diluted product was added to a mixture of 9 µl of Hidi (Hi-Di Formamide for 3500 Dx/3500xl Dx) + Liz500 (GeneScan 500 LIZ Size Standard), and the ratio of Hidi to Liz500 was 1,000:15. The amplicon length of each simple sequence repeat (SSR) was determined with an ABI 3500xl Genetic Analyzer (Applied Biosystems, Waltham, MA). The length of the fragment was scored in GeneMarker version 2.7.0 (SoftGenetics, State College, PA).

Data analyses. *Genetic diversity.* The samples were assigned to 12 subpopulations based on their geographic location and sampling time. The samples from the Sichuan and Hubei regions were divided into seven subpopulations: 2017_CB, 2017_CN, 2018_CB, 2017_EB, 2017_EN, 2018_EB, and 2018_EN. The other five subpopulations were 2017_GS, 2017_YN, 2017_GZ, 2018_YN, and 2018_GS (Supplementary Table S1). A dataset of the samples containing all the allele sizes of each locus was managed in Microsoft Excel according to the GenAlEx 6.5 data format (Peakall and Smouse 2012).

POPPR (version 2.5.0) (Kamvar et al. 2014), an R package, was used to explore genetic diversity. To assess whether the number of SSR loci was able to discriminate all the different individuals, the genotype accumulation curve (the default is $2/N$, where N equals the number of isolates in the dataset) was plotted. The locus-based statistic was used to calculate indices including the number of alleles, evenness, and the ratio of missing data to all samples. The POPPR function *info table* was used to discard the loci with >10% missing data for the following analysis. A search for shared multilocus genotypes (MLGs) between each pair of subpopulations was performed. If an MLG was repeated in multiple subpopulations, then the MLG was inferred to have migrated (Hovmøller et al. 2008). Genotypic richness was measured by the number of expected MLGs, which approximates the number of genotypes that would be expected for the largest shared sample size based on rarefaction, and the shared sample size should be ≥10. Next, we calculated genotypic evenness, which equaled the number of MLGs divided by population size and represented the distribution of genotype abundance (Grünwald et al. 2003). Simpson's index (Simpson 1949), corrected Simpson's index, and Shannon's information index (Shannon 2001) were also calculated, which reflected population richness and uniformity.

Population relationships and migration patterns. An analysis of molecular variance was performed in GenAlEx 6.5 with 9,999 permutations to detect genetic differences in the given subpopulation (Excoffier et al. 1992). To estimate the genetic differentiation between all subpopulations, pairwise F_{ST} was calculated based on heterozygote deficiency in both subpopulations. In addition, gene flow Nm ($Nm = [(1/F_{ST}) - 1]/4$) between each pair of subpopulations was calculated based on the migration rate. A greater value indicates more frequent gene flow. A high level of gene flow between populations is generally inferred when Nm is >4, an Nm value between 1 and 4 indicates a considerable amount of gene flow between populations, and an Nm value <1 implies low gene flow between populations (Slatkin 1989).

To infer population subdivision, we performed an analysis based on the Bayesian method in STRUCTURE 2.3.4 (Pritchard et al. 2000) to estimate the population structure and admixture extent of all populations. All samples were analyzed by K clusters ranging from one to 14 based on genotypes. Markov chain Monte Carlo replications were run for 40,000 iterations, with a 10,000 burn-in period for each K with 15 independent runs. The result from STRUCTURE was uploaded to STRUCTURE HARVESTER (http://taylor0.biology.ucla.edu/structureHarvester) to obtain the best K value based on the maximized rate of change in log likelihood values (ΔK) (Evanno et al. 2005).

In addition, multivariate discriminant analysis of principal components (DAPC) was performed with the *dapc ()* function, which is part of the *adegenet* package in the R environment (Jombart et al. 2010). All isolate SSR data were subjected to DAPC after clone correction. Meanwhile, to better distinguish the 2017 samples from the 2018 samples, the 2017 and 2018 samples were subjected to DAPC.

Air mass transport analysis with HYSPLIT. The monthly meteorological data (i.e., NCEP/NCAR 2.5° × 2.5° Reanalysis Data for the period from 1948 to present) collected four times a day (i.e., at 00, 06, 12, and 18 UTC) were used. Variables such as the u and v wind components, temperature, and humidity were obtained from the National Centers for Environmental Prediction and the National Center for Atmospheric Research (https://psl.noaa.gov/data/gridded/data.ncep.reanalysis.html). The HYSPLIT model was used to calculate the trajectories of air mass on the Yunnan–Guizhou Plateau at a height of 1,500 m above the ground from 30 December to 31 January of the following year over 30 years (from 1990 to 2020). We hypothesized that the spores of *Pst* would be released at 00, 06, 12, and 18:00 UTC every day. We simulated transport continuously for 48 h to calculate the forward airflow trajectories. Analysis of air mass transport was performed via geographic information system techniques (MeteoInfoMap version 2.2.1, China) loaded into TrajStat 1.4.8 (geographic information system–based software) as an external process. Thereafter, all the trajectory data were combined, and the clustering function was used for dimensionality reduction, in which several kinds of mean trajectories were calculated by TrajStat.

Results

SSR markers. A total of 13 SSR markers were used in this study and produced 48 alleles, with the number of alleles per locus ranging from two (RJ3, CPS34, and CPS36) to seven (RJ24). The values of evenness were between 0.316 for RJ21 and 0.997 for RJ3, and the mean value was 0.638, showing high diversity in the population. Locus RJ3 (0.997) had the highest evenness value, indicating that the alleles occurred at nearly equal frequencies in all samples. The percentage of missing values was low and ranged from 0 to 6.39% (average 2.31%) (Table 1). The genotype accumulation curve saturated, indicating that the 13 loci used in this study can distinguish all genotypes among the isolates (Supplementary Fig. S1).

Genetic diversity of subpopulations. A total of 249 unique MLGs were identified in 12 subpopulations with the 13 primer pairs. Genotypic evenness is a measure of the distribution of genotypic abundance, where the values of populations dominated by a single genotype are close to 0, whereas equally abundant genotypes yield values equal to 1. In our study, the genotypic evenness value ranged

from 0.188 to 1, with an overall value of 0.418. The highest genotypic evenness was 1 in the 2017_GS population, indicating that this population had the highest proportion of unique symptoms in all subpopulations. The 2018_GS population had the highest genotypic evenness value among all 2018 populations. We calculated the Shannon–Wiener index and Simpson's index to measure genotypic diversity. Overall, the Shannon–Wiener index and Simpson's index indicated that the diversity of the 2018 population was higher than that of the 2017 population, probably because the number of samples of the 2018 populations was larger than that of the 2017 samples. In the 2017 population, the Gansu subpopulation had the highest genotypic evenness value, but Simpson's index and Shannon's index were not the highest. This finding might be an artifact of sample size, because there were fewer samples from Gansu than from other regions because of the influence of disease occurrence. However, the sample could represent the population from the Gansu area because the sampling location covered a wide distribution there. To eliminate the effects of sample size, we calculated the corrected Simpson's index by multiplying the value of Simpson's index by $N/(N - 1)$. The 2017_GS population showed the highest genetic diversity (value equal to 1), and the 2017_EN population had the lowest genetic diversity (value equal to 0.425) (Table 2).

Population differentiation and relationships between pairs. Analysis of molecular variance showed that the majority of the variation occurred within subpopulations (69%, $P = 0.001$), and similar rates of variation were found between subpopulations (15%, $P = 0.001$) and between samples within subpopulations (16%, $P = 0.001$) (Table 3).

In addition, the F_{ST} and Nm between 12 subpopulations were used to analyze pairwise genetic differentiation. Among the seven subpopulations in 2017 (Table 4), 2017_YN and 2017_GZ had the lowest F_{ST} value (0.005) and the highest gene flow (45.381), indicating that they may have originated from the same ancestral population. Five subpopulations (2017_YN, 2017_GZ, 2017_CN, 2017_EN, and 2017_EB) showed higher gene flow than other subpopulations (Nm >4, F_{ST} <0.05). Although the Nm value (1.163) between 2017_CB and 2017_EB was >1, these subpopulations showed a low level of gene flow, which indicated indistinctive gene flow levels between 2017_GS and 2017_EN, 2017_GS and 2017_EB, 2017_CB and 2017_EN, and 2017_CB and 2017_EB (Nm <1, FST >0.1). Subpopulations 2017_GS and 2017_CB had higher levels of gene flow (Nm >4, F_{ST} <0.05). These results indicated that the inoculum was more frequently exchanged between the Hubei (2017_EN and 2017_EB) region and southwestern China than

Table 1. Primer information for the 13 simple sequence repeat markers used in this study

Locus	Sequence of primer (5′–3′)	Motif	Annealing temperature (°C)	Size range (bp)	Number of alleles	Evenness	% Missing	Reference
RJ3	GCAGCACTGGCAGGTGG GATGAATCAGGATGGCTCC	$(TGG)_8$	61	203–205	2	1.00	2.18	Enjalbert et al. (2002)
RJ4	GTGGGTTGGGCTGGAGTC GCTAATCCATTCCACGCACC	$(TGG)_8$	58	202–208	3	0.42	2.52	Enjalbert et al. (2002)
RJ18	CTGCCCATGCTCTTCGTC GATGAAGTGGGTGCTGCTG	$(TGT)_5$	59	334–361	4	0.64	6.22	Enjalbert et al. (2002)
RJ20	AGAAGATCGACGCACCCG CCTCCGATTGGCTTAGGC	$(CAG)_4$	56	284–292	3	0.90	3.03	Enjalbert et al. (2002)
RJ21	TTCCTGGATTGAATTCGTCG CAGTTCTCACTCGGACCCAG	$(GTT)_6$	55	164–179	3	0.32	0.34	Enjalbert et al. (2002)
RJ24	TTGCTGAGTAGTTTGCGGTGAG CTCAAGCCCATCCTCCAACC	$(GTT)_{5+9}$	58	274–301	7	0.78	5.55	Enjalbert et al. (2002)
RJ27	CGTCCCGACTAATCTGGTCC ATGAGTTAGTTTAGATCAGGTCGAC	$(TC)_{10}$	52	228–244	4	0.64	1.68	Enjalbert et al. (2002)
CPS08	GATAAGAAACAAGGGACAGC CAGTGAACCCAATTACTCAG	$(CAG)_{14}$	56	186–213	6	0.63	0.34	Chen et al. (2009)
CPS09	CGGGAGAAGACCTGAGC AGAAAACGGAATGTAATGTG	$(GTT)_9$	55	239–260	4	0.75	6.39	Chen et al. (2009)
CPS10	TCTACTGGGCAGACTGGTC CGGTTTGTTTTGTCGTTTC	$(TAG)_8$	56	321–333	3	0.34	1.01	Chen et al. (2009)
CPS15	GATGGGGAAAAGTAAGAAGT GGTGGGGGATGTAAGTATGTA	$(ATG)_5$	56	218–227	5	0.64	0.67	Chen et al. (2009)
CPS34	GTTGGCTACGAGTGGTCATC TAACACTACACAAAAGGGGTC	$(TTTGG)_4$	56	105–110	2	0.57	0	Chen et al. (2009)
CPS36	TCCAGGCAGTAAATCAGACGC ATCAGCAGGTGTAGCCCCATC	$(CTCTAG)_3$	58	128–131	2	0.43	0.17	Chen et al. (2009)
Total/mean					48	0.64	2.31	

Table 2. Genetic diversity in *Puccinia striiformis* f. sp. *tritici* subpopulations based on 13 simple sequence repeat loci

Subpopulation	No. of samples	No. of multilocus genotypes	Genotypic evenness	Nei's index	Simpson's index	Corrected Simpson's index	Shannon's index
2017_YN	89	41	0.46	0.27	0.90	0.91	3.04
2017_GZ	30	17	0.57	0.25	0.85	0.88	2.41
2017_CN	16	7	0.44	0.16	0.65	0.69	1.45
2017_EN	16	3	0.19	0.16	0.40	0.43	0.70
2017_EB	31	11	0.36	0.20	0.71	0.73	1.68
2017_GS	7	7	1.00	0.28	0.86	1.00	1.95
2017_CB	15	13	0.87	0.30	0.92	0.98	2.52
2018_CB	124	61	0.49	0.31	0.97	0.98	3.85
2018_GS	31	23	0.74	0.33	0.94	0.97	3.00
2018_EB	33	24	0.73	0.30	0.95	0.98	3.06
2018_EN	102	33	0.32	0.28	0.86	0.87	2.60
2018_YN	101	59	0.58	0.25	0.95	0.96	3.48
Total	595	249	0.42	0.32	0.97	0.97	4.65

between the Hubei region and northwestern China in 2017. Among the five subpopulations in 2018 (Table 5). A high level of gene flow was present among three of five subpopulations: 2018_GS, 2018_CB, and 2018_EB (Nm >4, FST <0.05). A large amount of gene flow occurred between these three subpopulations and the 2018_EN population (1 < Nm <4, 0.05 < F_{ST} <0.1). In addition, a high level of gene flow was present between 2018_YN and 2018_EN (Nm = 18.04), showing that the inoculum of southern Hubei in 2018 was probably from the Yunnan region. Although the Nm value was 4.309 between 2018_YN and 2018_EB, it was lower than that between 2018_GS and 2018_EB (Nm = 5.739). Thus, the inoculum of the northern Hubei region in 2018 might have originated mainly from the Gansu region and secondarily from the southwest of China through of southern Hubei.

The model-based clustering method implemented in the STRUCTURE analysis showed the strongest support for four clusters

Table 3. Analysis of molecular variance within and between *Puccinia striiformis* f. sp. *tritici* populations[a]

Source	df	SS	MS	Estimated variance	%	*P*
Between subpopulations	11	367.133	33.376	0.333	15%	0.001
Between isolates within subpopulations	583	1,333.002	2.286	0.363	16%	0.001
Within subpopulations	595	928.500	1.561	1.561	69%	0.001
Total	1,189	2,628.635		2.257	100%	

[a] df, degrees of freedom; MS, mean squares; SS, sum of squares.

Table 4. Pairwise genetic differentiation (under diagonal) and gene flow (above diagonal) between seven *Puccinia striiformis* f. sp. *tritici* subpopulations in 2017

Population	2017_YN	2017_GZ	2017_CN	2017_EN	2017_EB	2017_GS	2017_CB
2017_YN		45.381	4.748	5.797	7.833	1.255	1.550
2017_GZ	0.005[a]		5.759	6.505	8.821	1.249	1.400
2017_CN	0.050[a]	0.042[a]		6.273	8.001	1.080	1.102
2017_EN	0.041[a]	0.037[a]	0.038[a]		11.311	0.770	0.803
2017_EB	0.031[a]	0.028[a]	0.030[a]	0.022[a]		0.975	1.163
2017_GS	0.166[a]	0.167[a]	0.188[a]	0.245[a]	0.204[a]		6.035
2017_CB	0.139[a]	0.152[a]	0.185[a]	0.237[a]	0.177[a]	0.040[a]	

[a] Significant at $P < 0.01$.

Table 5. Pairwise genetic differentiation (under diagonal) and gene flow (above diagonal) between five *Puccinia striiformis* f. sp. *tritici* subpopulations in 2018

Population	2018_CB	2018_GS	2018_EB	2018_EN	2018_YN
2018_CB		26.843	7.213	2.542	2.977
2018_GS	0.009[a]		5.739	2.714	3.417
2018_EB	0.033[a]	0.042[a]		3.253	4.309
2018_EN	0.090[a]	0.084[a]	0.071[a]		18.040
2018_YN	0.077[a]	0.068[a]	0.055[a]	0.014[a]	

[a] Significant at $P < 0.01$.

Fig. 2. Assignment of *Puccinia striiformis* f. sp. *tritici* isolates to different genotype clusters in STRUCTURE software. Each line represents an isolate and is partitioned into the *K* colored segments that represent the probabilities of isolates belonging to different clusters.

(K = 4) (Fig. 2; Supplementary Fig. S2). For the seven subpopulations of 2017, cluster k4.3 (blue) were shared in five subpopulations (2017_YN, 2017_GZ, 2017_CN, 2017_EN, and 2017_EB) but not the 2017_GS and 2017_CB subpopulations. The samples from the 2017_GS and 2017_CB subpopulations were clustered in k4.4 (violet). For the five subpopulations of 2018, cluster k4.2 (green) was shared in only two subpopulations (2018_EN and 2018_YN), and cluster k4.4 (violet) was composed mainly of 2018_CB and 2018_GS. Subpopulation 2018_EB belonged mainly to clusters k4.3 (blue) and k4.4 (violet). DAPC (Fig. 3) confirmed the STRUCTURE results. DAPC was performed on the samples collected in 2017 and 2018. The 2017 group could be divided into two parts (Fig. 3A). Part 1 was distributed on the left (including five subpopulations, 2017_YN, 2017_GZ, 2017_CN, 2017_EN, and 2017_EB) and included subpopulations that have frequently exchanged genes. Part 2 was distributed on the right (including two subpopulations, 2017_GS and 2017_CB) and included subpopulations that have frequently exchanged genes. Part 1 and part 2 exhibited almost no exchange. In group 2018, subpopulation 2018_EB was clustered with the 2018_CB and 2018_GS subpopulations. Subpopulation 2018_EN was clustered with the 2018_YN subpopulation, and little exchange occurred between the two clusters (Fig. 3B).

In summary, all analyses strongly suggested that the 2017 Hubei population (2017_EB and 2017_EN) was probably from southwestern China (2017_YN, 2017_GZ, and 2017_CN). However, in the 2018 Hubei region, the northern Hubei population (2018_EB) was probably from northwestern China (2018_GS and 2018_CB), and the southern Hubei population (2018_EB) was probably from southwestern China (2018_YN).

Trajectory analysis. The analyzed upper airflow trajectory of the Yunnan–Guizhou Plateau at a 1,500 m height showed that seven upper airflow trajectories remained after dimensionality reduction. The trajectories fell into four orientations (Fig. 4). The mean type of the trajectories moved toward the northeast past the Jianghan Plain (accounting for 61.8% of the trajectories). These regions covered the central and eastern wheat acreage in China. The second type of trajectory moved toward the east (accounting for 11.64% of the trajectories). Wheat is seldom grown in these areas, such as Guangxi, Guangdong, and Jiangxi provinces. The third type moved from the Sichuan Basin to the Yunnan–Guizhou Plateau (accounting for 12.29% of the trajectories). The last type of trajectory moved toward Southeast Asia (accounting for 14.28% of the trajectories).

Discussion

Located in central China, Hubei has diverse topographies and landforms, covering the western mountainous region, southern Jianghan Plain, northeastern low-altitude hill region, and southeastern low-altitude hill region. The Yangtze River flows from west to east, and the Jianghan Plain and northern Hubei region are divided by Dahong Mountain and connected by the Hanshui Valley. Wheat is planted from late September to mid-October, and stripe rust appears

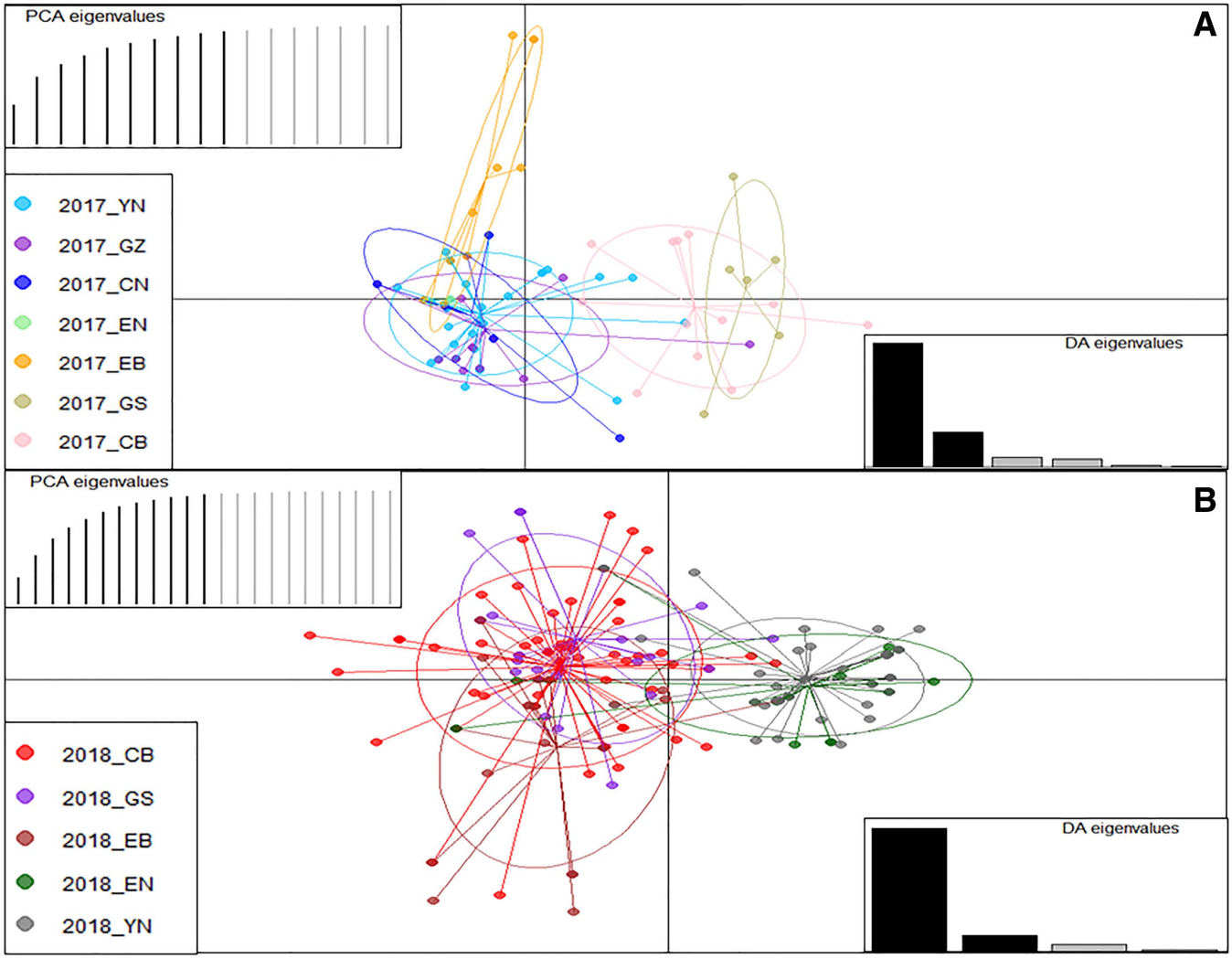

Fig. 3. Discriminant analysis of principal components for genotypes of *Puccinia striiformis* f. sp. *tritici* from different subpopulations and times based on 13 pairs of simple sequence repeat primers. **A,** All samples were collected in 2017, and seven subpopulations were defined. **B,** All samples were collected in 2018, and five subpopulations were defined. DA, discriminant analysis; PCA, principal component analysis.

in late February under natural conditions. *Pst* urediniospores cannot oversummer in the plain region of Hubei because the weather is too hot and humid, and the initial *Pst* inoculum of the plain region is probably from other regions (Chen et al. 2013). Previous studies suggested that the inoculum of *Pst* in Hubei might be from northwestern China. Early researchers speculated that the inoculum source of *Pst* from the Yunnan–Guizhou Plateau can spread to Hubei by upper airflow, without evidence (Brown and Hovmøller 2002). In 2017, wheat stripe rust caused a severe epidemic in China, which was characterized by early occurrence, fast and widespread, and serious damage. The disease first appeared in Hubei in late December 2016 and developed rapidly in Jianghan Plain and northwestern Hubei in late January 2017. In addition, the disease occurred at low levels in Gansu but at high levels in the Yunnan–Guizhou Plateau. Thus, the source of inoculum in Hubei was probably from other regions. Epidemic years provide the best opportunity to study disease epidemiology (Huang et al. 2018a). To explore the origin source areas of the Hubei inoculum in 2017, we collected samples from Hubei, Yunnan–Guizhou, Gansu, and Sichuan and genotyped with SSR markers. The F_{ST}, Nm, STRUCTURE, and DAPC results clearly showed that the *Pst* population from Hubei had low differentiation from and high gene flow with the populations from the Yunnan–Guizhou Plateau. This finding is inconsistent with previous reports (Chen et al. 2013). Thus, we collected 391 samples from four provinces (Yunnan, Sichuan, Gansu, and Hubei) in 2018. As revealed by the SSR analysis, the southern Hubei population had low genetic differentiation from the population from Yunnan, whereas the northern Hubei population had low genetic differentiation from the populations from Yunnan and Gansu. Thus, we suggest that the *Pst* inoculum of Hubei can be from two sources. One is from the south of Gansu, and the other is from the Yunnan–Guizhou Plateau. Usually, the southern part of Gansu is regarded as the most important inoculum source of *Pst* in China. The inoculum spread to Hubei through the upper air, reached the Hanshui Valley, and accumulated on wheat. Consequently, it infects wheat in eastern China (Wan et al. 2007; Zeng and Luo 2006). However, the inoculum of Hubei could also have originated from the Yunnan–Guizhou Plateau. The inoculum of the Yunnan–Guizhou region could spread to southern Hubei by upper airflow and then disperse to northern Hubei after multiplication on wheat.

The Yunnan–Guizhou Plateau is located in southwestern China and east of the Himalayan region (Nepal and Pakistan). The Himalayan region is proposed as the origin center of *Pst* worldwide (Ali et al. 2014). *Pst* urediniospores could be carried by wind along the south slope of the Himalayas to the Yunnan–Guizhou Plateau. In the present study, the *Pst* populations of the Yunnan–Guizhou region had higher levels of genetic diversity than those from other regions. In addition to the inoculum source from the Himalayas and strong ultraviolet rays at high altitudes, sexual recombination in local fields could also be important. Since the first report of an alternate host for *Pst* (Jin et al. 2010), 17 *Berberis* species susceptible to *Pst* have been identified in the Yunnan–Guizhou Plateau (Li et al. 2021b; Zhao et al. 2013). Natural *Pst* infections on plants of two barberry species were identified (Li et al. 2021b). Meanwhile, *Berberis* species infected with rust pathogens were observed in a field adjacent to wheat fields infected with wheat stripe rust (Li et al. 2021a).

The Yunnan–Guizhou Plateau has a varied topography and a cool climate (the annual average temperature is 5 to 24°C). Because of the climate and farming systems, wheat or volunteer plants can remain in fields all year long, providing green bridges for *Pst* throughout the year. In this area, wheat is planted from late October to early November and harvested from early May to mid-May of the following year. Stripe rust occurs throughout the winter because of the mild temperatures, and the spread of *Pst* urediniospores from the Yunnan–Guizhou region to other regions by airflow in winter is an important prerequisite for epidemics in these regions.

Long-distance transport is an important pathogen dispersal mode, particularly for urediniospores of *Pst*, which can be transmitted over a long distance by upper airflow. *Pst* could spread from western oversummering regions to eastern wheat belts of China in the 3 km of upper air (Xie et al. 1993). Chinese scientists analyzed 8-year trajectory data from the Yunnan–Guizhou Plateau, and the results showed that airflow can arrive at the middle and lower reaches of the Yangtze River from 1 October to 31 May (Li et al. 2021a). This indicates that *Pst* from the Yunnan–Guizhou Plateau is critical for the onset of this disease in other wheat planting regions. We found

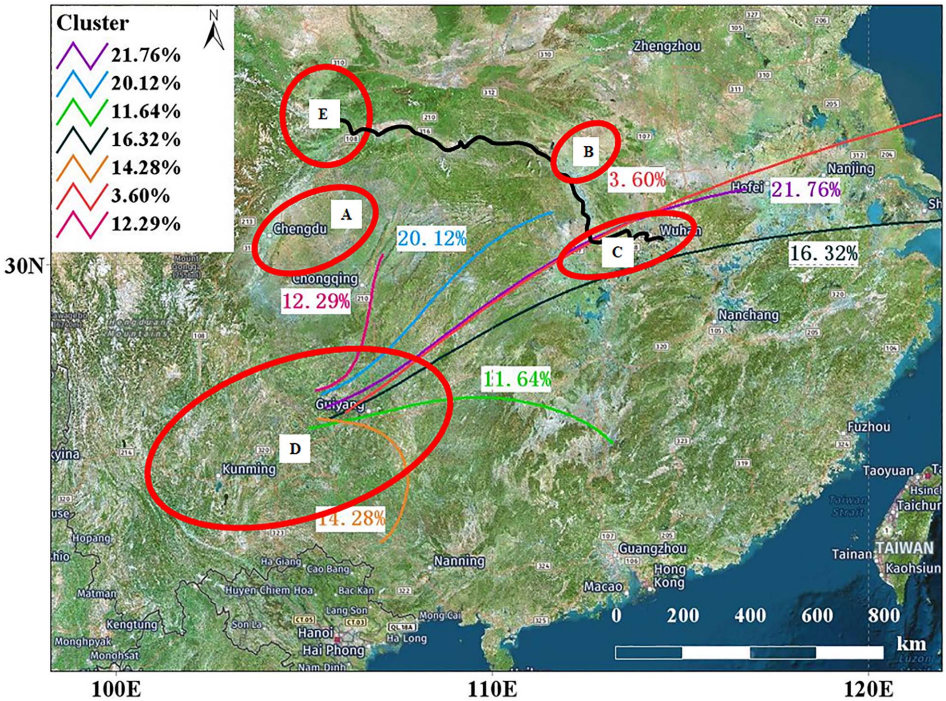

Fig. 4. Air trajectory of the Yunnan–Guizhou Plateau from 31 November to 31 January of the following year from 1990 to 2020 after dimensionality reduction. All trajectories were clustered into four orientations: north (pink), northeast (blue, violet, red, and black), east (green), and south (orange). Regions circled on the map: A, Sichuan Basin; B, northwestern Hubei region; C, Jianghan Plain; D, Yunnan–Guizhou Plateau; E, southeastern Gansu region; Black curve: Hanshui Valley.

the probability that the upper airflow from the Yunnan–Guizhou Plateau blew to the Jianghan Plain. Moreover, many valleys run from southwest to northeast between the Yunnan–Guizhou Plateau and the Jianghan Plain. This provides an advantage to *Pst* urediniospores, carrying them by upper airflow from the Yunnan–Guizhou Plateau to the Jianghan Plain along the valleys.

The source of *Pst* in the Huang–Huai wheat region is previously thought to be mainly northwestern Hubei (Wang et al. 2020), but the source of inoculum in the Jianghan Plain has not been well studied. After oversummering, urediniospores from oversummering regions in northwestern China, particularly southeastern Gansu, could be transported to northwestern Hubei in autumn along the upstream Hanshui River Valley by winds from northwest to southeast. Urediniospores can overwinter and multiply on wheat in northwestern Hubei and further spread to eastern China in spring (Chen et al. 2014). However, the inoculum from the Yunnan–Guizhou Plateau can spread directly to the Jianghan Plain because of the topography and airflow direction. Urediniospores migrate to the north of Hubei along the Hanshui Valley after accumulating on wheat in the Jianghan Plain. For virulent strains of *Pst*, the earlier the pathogen infects, the larger the clonal population size will be (Hovmøller et al. 2002). This might explain why the population of northwestern Hubei had a closer genetic relationship with the Gansu population than with the Yunnan–Guizhou population.

In the present study, we identified a new route of *Pst* dissemination in China. This information is important for understanding stripe rust epidemiology. The results of this study can help strengthen integrated stripe rust management in China.

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